Лабораторная работа №3

Настройка DHCP-сервера

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дисциплина: **Администрирование сетевых подсистем**

Цель работы

Приобретение практических навыков по установке и конфигурированию DHCP-сервера.

{C:\work\frank taleubou\vagrant} - Far 3.0.6648.0 x64 Administrator

```
C:\work\frank taleubou\vagrant                    C:\Program Files\Far Manager    13:26
n        Name          Name                      n        Name          Name
..                                                ..                      FarSky.lng
.vagrant                                          Addons                  FarSpa.lng
provision                                         Documentation           FarUkr.hlf
Makefile                                          Encyclopedia            FarUkr.lng
Vagrantfile                                       Plugins                 File_id.diz
                                                  VisualElements          lua51.dll
                                                  changelog               luafar3.dll
                                                  Far.exe                 luafar3.map
                                                  Far.exe.example.ini     RestoreOldPluginSetting}
                                                  Far.map                 SaveOldPluginSettings.c}
                                                  Far.VisualElementsMani} sqlite3.dll
                                                  FarBel.lng              sqlite3.map
                                                  FarCze.hlf
                                                  FarCze.lng
                                                  FarEng.hlf
                                                  FarEng.lng
                                                  FarGer.hlf
                                                  FarGer.lng
                                                  FarHun.hlf
                                                  FarHun.lng
                                                  FarIta.lng
                                                  FarLit.lng
                                                  FarPol.hlf
                                                  FarPol.lng
                                                  FarPor.lng
                                                  FarRus.hlf
                                                  FarRus.lng
                                                  FarSky.hlf
Makefile                  3316 18.10.25 01:22     ..                      Up    25.02.26 13:28
Bytes: 6,30 K, files: 2, folders: 2               Bytes: 19,6 M, files: 34, folders: 5
C:\work\frank taleubou\vagrant>
1Help  2UserMn 3View  4Edit  5Copy  6RenMov 7MkFold 8Delete 9ConfMn 10Quit  11Plugin
```

root@server:~

```
[vagrant@server ~]$ sudo -i
[root@server ~]# dnf -y install kea
Last metadata expiration check: 0:16:46 ago on Wed 04 Mar 2026 03:29:26 PM UTC.
Dependencies resolved.
================================================================================
 Package                       Arch      Version            Repository    Size
================================================================================
Installing:
 kea                           x86_64    2.6.4-1.el9        epel         1.3 M
Installing dependencies:
 kea-libs                      x86_64    2.6.4-1.el9        epel         3.0 M
 log4cplus                     x86_64    2.0.5-15.el9       epel         337 k
 mariadb-connector-c           x86_64    3.2.6-1.el9_0      appstream    195 k
 mariadb-connector-c-config    noarch    3.2.6-1.el9_0      appstream    9.8 k
 postgresql-private-libs       x86_64    13.23-1.el9_7      appstream    140 k

Transaction Summary
================================================================================
Install  6 Packages

Total download size: 5.0 M
Installed size: 18 M
Downloading Packages:
(1/6): log4cplus-2.0.5-15.el9.x86_64.rpm                     1.8 MB/s | 337 kB     00:00
(2/6): kea-libs-2.6.4-1.el9.x86_64.rpm                       4.9 MB/s | 3.0 MB     00:00
(3/6): mariadb-connector-c-3.2.6-1.el9_0.x86_64.rpm          443 kB/s | 195 kB     00:00
(4/6): kea-2.6.4-1.el9.x86_64.rpm                            1.9 MB/s | 1.3 MB     00:00
(5/6): mariadb-connector-c-config-3.2.6-1.el9_0.noarch.rpm    59 kB/s | 9.8 kB     00:00
(6/6): postgresql-private-libs-13.23-1.el9_7.x86_64.rpm      868 kB/s | 140 kB     00:00
--------------------------------------------------------------------------------
Total                                                         42 kB/s | 5.0 MB     02:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                          1/1
  Installing       : postgresql-private-libs-13.23-1.el9_7.x86_64             1/6
  Installing       : mariadb-connector-c-config-3.2.6-1.el9_0.noarch          2/6
  Installing       : mariadb-connector-c-3.2.6-1.el9_0.x86_64                 3/6
```

Рис. ины server.

```
[   44.621644] vmwgfx 0000:00:02.0: [drm] DMA map mode: Caching DMA mappings.
[   44.624452] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 262144 kB
, FIFO = 2048 kB, surface = 262144 kB
[   44.624462] vmwgfx 0000:00:02.0: [drm] MOB limits: max mob size = 131072 kB,
max mob pages = 262144
[   44.624467] vmwgfx 0000:00:02.0: [drm] Max GMR ids is 8192
[   44.624470] vmwgfx 0000:00:02.0: [drm] Max number of GMR pages is 1048576
[   44.624472] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 262144
kiB
[   44.743257] vmwgfx 0000:00:02.0: [drm] Screen Target display unit initialized
[   44.749736] vmwgfx 0000:00:02.0: [drm] Fifo max 0x00200000 min 0x00001000 cap
 0x00000355
[   44.750680] vmwgfx 0000:00:02.0: [drm] Using command buffers with DMA pool.
[   44.750704] vmwgfx 0000:00:02.0: [drm] Available shader model: SM_5.
[   44.754230] [drm] Initialized vmwgfx 2.20.0 20211206 for 0000:00:02.0 on mino
r 0
[   44.770198] fbcon: vmwgfxdrmfb (fb0) is primary device
[   44.786079] Console: switching to colour frame buffer device 160x50
[   44.800855] vmwgfx 0000:00:02.0: [drm] fb0: vmwgfxdrmfb frame buffer device
[   44.813079] snd_intel8x0 0000:00:05.0: intel8x0_measure_ac97_clock: measured 56364 usecs (2640 samples)
[   44.813091] snd_intel8x0 0000:00:05.0: clocking to 49190
[   50.307840] systemd-rc-local-generator[739]: /etc/rc.d/rc.local is not marked executable, skipping.
[   53.185485] NET: Registered PF_QIPCRTR protocol family
[   58.765459] e1000: eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
[   58.808401] IPv6: ADDRCONF(NETDEV_CHANGE): eth0: link becomes ready
[   58.900760] e1000: eth1 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
[   58.965799] Warning: Unmaintained driver is detected: ip_set
[   59.814644] IPv6: ADDRCONF(NETDEV_CHANGE): eth1: link becomes ready
[   91.092689] block sda: the capability attribute has been deprecated.
```

Остальные примеры задания конфигураций подсетей удалите.

3. Настройте привязку dhcpd к интерфейсу eth1 виртуальной машины server:


```
"interfaces-config": {
  "interfaces": [ "eth1" ]
},
```

4. Проверьте правильность конфигурационного файла:

```
kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
```

```
// DHCPv4 configuration starts here. This section will be read by DHCPv4 server
// and will be ignored by other components.
"Dhcp4": {
    // Add names of your network interfaces to listen on.
    "interfaces-config": {
        // See section 8.2.4 for more details. You probably want to add just
        // interface name (e.g. "eth0" or specific IPv4 address on that
        // interface name (e.g. "eth0/192.0.2.1").
        "interfaces": ["eth1" ]

        // Kea DHCPv4 server by default listens using raw sockets. This ensures
        // all packets, including those sent by directly connected clients
        // that don't have IPv4 address yet, are received. However, if your
        // traffic is always relayed, it is often better to use regular
        // UDP sockets. If you want to do that, uncomment this line:
        // "dhcp-socket-type": "udp"
    },

    // Kea supports control channel, which is a way to receive management
    // commands while the server is running. This is a Unix domain socket that
    // receives commands formatted in JSON, e.g. config-set (which sets new
    // configuration), config-reload (which tells Kea to reload its
    // configuration from file), statistic-get (to retrieve statistics) and many
    // more. For detailed description, see Sections 8.8, 16 and 15.
    "control-socket": {
        "socket-type": "unix",
        "socket-name": "kea4-ctrl-socket"
    },

    // Use Memfile lease database backend to store leases in a CSV file.
    // Depending on how Kea was compiled, it may also support SQL databases
```

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```
(gedit:27131): dconf-WARNING **: 19:54:00.846: failed to commit chang
xecute child process “dbus-launch” (No such file or directory)
[root@server ~]#  /var/named/master/fz
-bash: /var/named/master/fz: No such file or directory
[root@server ~]# cd  ..
[root@server /]# cd  /var/named/master/fz
-bash: cd: /var/named/master/fz: No such file or directory
[root@server /]# cd /var
[root@server var]# cd master
-bash: cd: master: No such file or directory
[root@server var]# systemctl restart named
Failed to restart named.service: Unit named.service not found.
[root@server var]# ping dhcp.user.net
ping: dhcp.user.net: Name or service not known
[root@server var]# chown -R named:named /var/named
chown: invalid user: ‘named:named’
[root@server var]#
```

```
    client.vm.provision "client routing",
                type: "shell",
                preserve_order: true,
                run: "always",
                path: "provision/client/01-r

    client.vm.provider :virtualbox do |v|
        v.linked_clone = true
# Customize the amount of memory on the VM
        v.memory = 4000
```

```
xecute child process "dbus-launch" (No such file or directory)
[root@server /]# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe69:a8d  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:a00:27ff:fe69:a8d  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:69:0a:8d  txqueuelen 1000  (Ethernet)
        RX packets 1479236  bytes 2019630895 (1.8 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 178954  bytes 12543603 (11.9 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.1  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::a00:27ff:fee3:341f  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:e3:34:1f  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 44  bytes 5322 (5.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 17  bytes 2039 (1.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 17  bytes 2039 (1.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[root@server /]#
```

```
** (gedit:27370): WARNING **: 20:14:42.249: Set document metadata failed
etadata::gedit-spell-language not supported

** (gedit:27370): WARNING **: 20:14:42.249: Set document metadata failed
etadata::gedit-encoding not supported

** (gedit:27370): WARNING **: 20:14:43.803: Set document metadata failed
etadata::gedit-position not supported

(gedit:27370): dconf-WARNING **: 20:14:43.832: failed to commit changes
xecute child process “dbus-launch” (No such file or directory)
[root@server /]# named-checkconf
bash: named-checkconf: command not found...
Packages providing this file are:
'bind'
'bind9.18'
[root@server /]#
```

```
(gedit:27448): dconf-WARNING **: 20:18:21.001: failed to commit changes to dconf: Failed to e
xecute child process “dbus-launch” (No such file or directory)
[root@server /]# chown kea:kea /etc/kea/kea-dhcp-ddns.conf
[root@server /]# kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf
Syntax check failed with: /etc/kea/tsig-keys.json:6.1: Invalid character: "
[root@server /]# systemctl enable --now kea-dhcp-ddns.service
Created symlink /etc/systemd/system/multi-user.target.wants/kea-dhcp-ddns.service → /usr/lib/
systemd/system/kea-dhcp-ddns.service.
[root@server /]# systemctl status kea-dhcp-ddns.service
× kea-dhcp-ddns.service - Kea DHCP-DDNS Server
     Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disable>
     Active: failed (Result: exit-code) since Wed 2026-03-04 20:18:53 UTC; 14s ago
   Duration: 58ms
       Docs: man:kea-dhcp-ddns(8)
    Process: 27518 ExecStart=/usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf (code=ex>
   Main PID: 27518 (code=exited, status=1/FAILURE)
        CPU: 31ms

Mar 04 20:18:53 server.user.net systemd[1]: Started Kea DHCP-DDNS Server.
Mar 04 20:18:53 server.user.net kea-dhcp-ddns[27518]: 2026-03-04 20:18:53.091 INFO  [kea-dhc>
Mar 04 20:18:53 server.user.net kea-dhcp-ddns[27518]: 2026-03-04 20:18:53.095 FATAL [kea-dhc>
Mar 04 20:18:53 server.user.net kea-dhcp-ddns[27518]: Service failed: Could Not load configu>
Mar 04 20:18:53 server.user.net systemd[1]: kea-dhcp-ddns.service: Main process exited, code>
Mar 04 20:18:53 server.user.net systemd[1]: kea-dhcp-ddns.service: Failed with result 'exit->
lines 1-15/15 (END)
```

Вывод

В ходе выполнения лабораторной работы были приобретены практические навыки по установке и конфигурированию DHCP-сервера.

Спасибо за внимание!